Control Theory

Chap 6: Optimal and Robust Control

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Chap 6: Optimal and Robust Control

1. Introduction
2. Optimal Control and Hamiltion - Jabobian - Bellman Equation
3. Linear Quadratic Regulator
4. Robust Control
5. Chapter Conclusion

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Review

- Computation optimization based PID tuning approach.
- Open question: how to determine the feedback gain K.

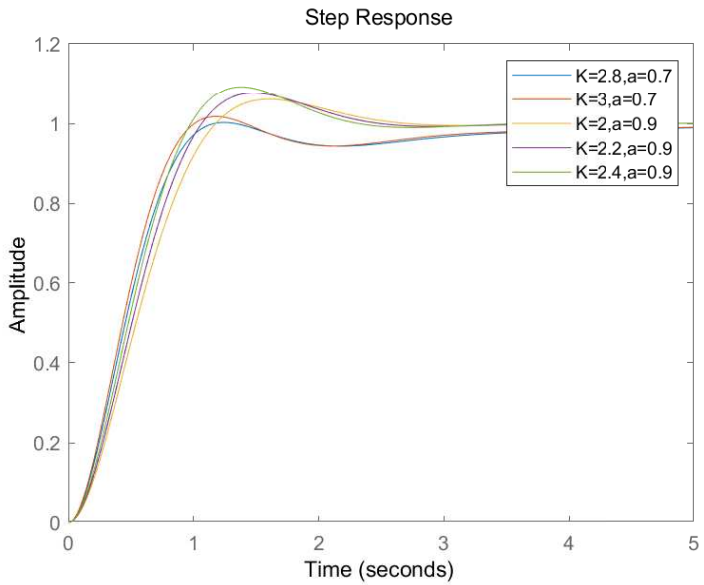


Figure 1: Responses under Different Gains

Select K of u

$$u = K^* x(t) \tag{1}$$

for the system

$$\dot{x} = Ax(t) + Bu(t) \tag{2}$$

Optimization

Doing a job in the best possible way. Optimal control problems arise in designing a controller to minimize a performance index. The key to using optimal control theory is to develop a method to tune the design parameters to achieve the desired performance and stability in the control system.

Mathematical Constitution of Optimal Control

A mathematical statement of the optimal control problem consists of

- A description of the system plant.
- A description of system constraints and possible alternatives.
- A description of the task to be accomplished.
- A criterion for judging optimal performance.



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Consider a dynamic system

$$\dot{x} = f\left(x, u, t\right), x(t_0) = x_0 \tag{3}$$

(optimality) cost performance index:

$$J = \int_{t_0}^{T} L\left(x\left(\tau\right), u\left(\tau\right), \tau\right) d\tau + S\left(x\left(T\right)\right) \tag{4}$$

Aim: Find an optimal control policy u^* can minimize the cost index J

$$J = J\left(x(t_0), u\left(\bullet\right), t_0\right) \tag{5}$$

Denote J^* the optimal cost under the optimal control policy u^*

$$\begin{array}{rcl} J^*\left(x_0, t_0\right) & = & \int_{t_0}^{T} L\left(x^*\left(\tau\right), u^*\left(\tau\right), \tau\right) d\tau + S\left(x^*, T\right) \\ & = & \min_{u[t_0,T]} J\left(x_0, u, t_0\right) \end{array} \tag{6}$$

It's a functional of the control input $u_{[t_0,T]}$, the initial state x_0 and time t_0:

$$\begin{array}{rcl} J^*(x_0, t_0) & = & \min_{u[t_0,T]} J(x_0, u(\bullet), t_0) \\ & = & \min_{u[t_0,T]} \left[\int_{t_0}^{T} L\left(x, u, \tau\right) d\tau + S\left(x\left(T\right)\right)\right] \end{array} \tag{7}$$



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If the system starts at an arbitrary initial condition x and time t, the optimal cost from x can be verified as

$$J^*(x,t) = \min_{u[t,T]} \left[\int_t^T L(x,u,\tau)d\tau + S(x(T)) \right] \tag{8}$$

also, it can be broken into divisions

$$J^*(x,t) = \min_{u[t,t_1]} \min_{u[t_1,T]} \left[\int_t^{t_1} L(x,u,\tau)d\tau + \int_{t_1}^T L(x,u,\tau)d\tau + S(x(T)) \right] \tag{9}$$

Principle of Optimality:

Whatever the initial state and initial decision are, the remaining decisions must constitute an optimal policy of the state resulting from the first decision. The optimal cost from x at time t to a terminal state $x(T)$ can be computed by minimizing the sum of

- (a) the cost to go from $x = x(t)$ to $x_1 = x(t_1)$;
- (b) the optimal cost from x_1 onwards.

$$J^*(x,t) = \min_{u_{t,t_1}} \left[\int_t^{t_1} L(x,u,\tau)d\tau + \underbrace{\min_{u_{t_1,T}} \int_{t_1}^{T} L(x,u,\tau)d\tau + S(x(T))}_{J^*(x,t_1)} \right] \tag{10}$$

$$J^*(x,t) = \min_{u_{t,t_1}} \left[\int_t^{t_1} L(x,u,\tau)d\tau + J^*(x,t_1) \right] \tag{11}$$

If one defines $t_1 = t + \Delta t$ (main idea)

$$J^*(x,t) = \min_{u_{t,t_1}} \left[\int_t^{t+\Delta t} L(x,u,\tau)d\tau + J^*(x(t+\Delta t), t+\Delta t) \right] \tag{12}$$

by the Tayler series, it has

$$J^*(x,t) = \min_{u_{t,t_1}} \left[L(x,u,\tau)\Delta t + J^*(x,t) + \left(\frac{\partial J^*(x,t)}{\partial x} \right)^T \Delta x + \frac{\partial J^*(x,t)}{\partial t}\Delta t + \underbrace{O(\Delta t^2)}_{H.O.T} \right] \tag{13}$$

and

$$0 = \min_{u_{t,t_1}} \left[L(x,u,\tau)\Delta t + \left(\frac{\partial J^*(x,t)}{\partial x} \right)^T \Delta x + \frac{\partial J^*(x,t)}{\partial t}\Delta t + \underbrace{O(\Delta t^2)}_{H.O.T.} \right] \tag{14}$$

where

$$\frac{\partial J^*}{\partial x}(x,t) = \left[\begin{array}{ccc} \frac{\partial J^*}{\partial x_1} & \cdots & \frac{\partial J^*}{\partial x_n} \end{array} \right] \in \Re^n$$

one has that

$$-\frac{\partial J^*(x,t)}{\partial t} = \min_{u} \left[L(x,u,\tau) + (\frac{\partial J^*(x,t)}{\partial x})^T \dot{x} \right] \tag{15}$$

by letting $\Delta t \to 0$

$$-\frac{\partial J^*(x,t)}{\partial t} = \frac{\min_{u_{t,t_1}} \left[L(x,u,\tau)\Delta t + (\frac{\partial J^*(x,t)}{\partial x})^T \Delta x + \underbrace{O(\Delta t^2)}_{H.O.T.} \right]}{\Delta t}$$



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The Hamiltonian then can be introduced

$$H(x, \delta_x J^*(x,t), u, t) = L(x, u, \tau) + \frac{\partial J^*(x,t)}{\partial x} f(x, u, t) \tag{16}$$

Hamilton - Jacobian - Bellman (HJB)

$$-\frac{\partial J^*(x,t)}{\partial t} = \min_u H(x, \nabla_x J^*(x,t), u, t) = H^*(x, \nabla_x J^*(x,t), t) \tag{17}$$

whose boundary condition, setting $t_0 = T$ in the cost index it yields

$$J^*(x(T), T) = S(x(T)) \tag{18}$$



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HJB Optimal Processing

- Dynamics:

$$\dot{x} = f(x,u), x(t_0) = x_0 \tag{19}$$

- Performance index:

$$J(x,u,t) = \int_t^T L(x,u,\tau)d\tau + S(x(T)) \tag{20}$$

- Optimal Cost:

$$J^* = \min_{u_{[t,T]}}[J] \tag{21}$$

- Hamiltonian:

$$H(x,u,t) = L(x,u,t) + \frac{\partial J^*(x,t)}{\partial x} f(x,u) \tag{22}$$

- Optimal control(Policy):

$$\nabla H_u(x,u,t) = 0 \Rightarrow u^* \Rightarrow H^*(x,u^*,t) \tag{23}$$

- HJB - Equation:

$$\begin{cases} -\frac{\partial J^*(x,t)}{\partial t} &= H^*(x, \nabla_x J^*(x,t), t) \\ J^*(x(T),T) &= S(x(T)) \end{cases} \tag{24}$$

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Example

Set up the HJB equation for the system below

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -2x_1 - 3x_2 + u \end{aligned} \tag{25}$$

where $x(0) = [1\ 2]^T$, with the performance index

$$J = \int_0^1 \underbrace{(x_1^4 + u^2)}_{L(x,u,t)}\, dt + \underbrace{(x_1^2(1) + x_2^2(1))}_{S(x(T))}$$

The Hamiltonian is

$$\begin{aligned} H(x, u, \nabla_x J^*, t) &= L(x, u, t) + \frac{\partial J^*}{\partial x} f(x, u) \\ &= x_1^4 + u_1^2 + \frac{\partial J^*}{\partial x} f(x, u) \\ &= x_1^4 + u_1^2 + \frac{\partial J^*}{\partial x_1}(x_2) + \frac{\partial J^*}{\partial x_2}(-2x_1 - 3x_2 + u) \end{aligned} \tag{26}$$

Minimize the Hamiltonian by differentiating the right-hand side of (26) to the control input u, equating the resulting derivative to zero, it has

Example

$$\left(\nabla H_u = 0 = 2u^* + \frac{\partial J^*}{\partial x_2}\right) \Rightarrow \left(u^* = -\frac{1}{2}\frac{\partial J^*}{\partial x_2}\right) \tag{27}$$

Substituting (27) into (26), it obtains

$$\begin{aligned} H(x, u, \nabla_x J^*, t) &= x_1^4 + u_1^2 + \frac{\partial J^*}{\partial x_1}(x_2) + \frac{\partial J^*}{\partial x_2}(-2x_1 - 3x_2 + u) \\ &= x_1^4 + \frac{1}{4}\frac{(\partial J^*)^2}{(\partial x_2)^2} + \frac{\partial J^*}{\partial x_1}(x_2) + \frac{\partial J^*}{\partial x_2}(-2x_1 - 3x_2 - \frac{1}{2}\frac{\partial J^*}{\partial x_2}) \end{aligned}$$

It has the HJB equation

$$\begin{aligned} -\frac{\partial J^*}{\partial t} &= H^*(x, \nabla_x J^*(x,t), t) \\ &= x_1^4 + \frac{1}{4}\frac{(\partial J^*)^2}{(\partial x_2)^2} + \frac{\partial J^*}{\partial x_1}(x_2) + \frac{\partial J^*}{\partial x_2}(-2x_1 - 3x_2 - \frac{1}{2}\frac{\partial J^*}{\partial x_2}) \\ &= x_1^4 - \frac{1}{4}\frac{(\partial J^*)^2}{(\partial x_2)^2} + \frac{\partial J^*}{\partial x_1}(x_2) - 2\frac{\partial J^*}{\partial x_2}x_1 - 3\frac{\partial J^*}{\partial x_2}x_2 \end{aligned} \tag{28}$$

the boundary condition $J^* = x_1^2(T) + x_2^2(T)$, where $T = 1$.



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3 Linear Quadratic Regulator
- Finite Horizon
- Infinite Horizon



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The linear quadratic regulator (LQR) is one of the most widely used control design methods for state-space system.
Consider the linear non-autonomous system

$$\dot{x} = A(t)x + B(t)u, x(t_0) = x_0$$

where $x \in \mathcal{R}^n, u \in \mathcal{R}^m$ with the quadratic performance index

$$J = \int_{t_0}^{T} (x^T Q x + u^T R u) d\tau + x^T(T) Q_T x(T) \tag{29}$$

where the cost weight matrices (Q,R,Q_T) are symmetric positive semi-definite, positive definite, and positive definite, respectively: i.e

$$Q = Q^T \geq 0, R = R^T > 0, Q_T = Q_T^T > 0,$$



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The LQR Hamiltonian is

$$H = x^T Q x + u^T R u + \frac{\partial J^*}{\partial x}(A(t)x + B(t)u) \tag{30}$$

the gradient of H with respect to u and equating it to zero products

$$\frac{\partial H}{\partial u} = 2Ru + B^T \nabla_x J^*(x,t) = 0 \tag{31}$$

the optimal control is

$$u^* = -\frac{1}{2} R^{-1} B^T \nabla_x J^*(x,t) \tag{32}$$

Substituting (32) into (30), it yields the HJB equation

$$-\frac{\partial J^*}{\partial t} = x^T Q x + \frac{1}{4}\frac{\partial J^*}{\partial x} B R^{-1} B^T \nabla_x J^* + \frac{\partial J^*}{\partial x}(A(t)x - B(t)\frac{1}{2} R^{-1} B^T \nabla_x J^*(x,t))$$



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We guess J^* is a quadratic time-varying function of the system state.

$$J^* = J^*(x(t), t) = x^T(t)P(t)x(t) \tag{33}$$

where $P(t)$ is positive definite and

$$\frac{\partial J^*(x,t)}{\partial t} = x^T \dot{P}(t)x, \nabla_x J^*(x,t) = 2P(t)x \tag{34}$$

It can verify that

$$x^T \left[-\dot{P}(t) - P(t)A - A^T P - Q + P(t)BR^{-1}B^T P(t)\right] x = 0 \tag{35}$$

with the boundary condition $P(T) = Q_T$, which is satisfied for any state x, the following initial value problem must be true:

$$u^*(x,t) = -\tfrac{1}{2}R^{-1}B^T \frac{\partial J^*(x,t)}{\partial x} = -\underbrace{R^{-1}B^T P(t)}_{K(t)} x = -K(t)x \tag{36}$$

Steps for Finite Horizon Optimal Control

- Dynamics

$$\dot{x} = A(t)x + B(t)u, x(t_0) = x_0 \tag{37}$$

- Performance index

$$J = \int_{t_0}^{T} (x^T Q x + u^T R u) d\tau + x^T(T) Q_T x(T) \tag{38}$$

- Riccati Equation

$$\left[\begin{array}{rcl} -\dot{P}(t) & = & P(t)A + A^T P(t) + Q - P(t)BR^{-1}B^T P(t) \\ P(T) & = & Q_T \end{array} \right] \tag{39}$$

- Optimal Control

$$u^* = -R^{-1}B^T P(t)x = -K(t)x \tag{40}$$

- Closed - Loop System

$$\dot{x} = (A(t) - B(t)K(t))x, x(0) = x_0 \tag{41}$$

3 Linear Quadratic Regulator
- Finite Horizon
- Infinite Horizon

Consider the quadratic performance index on an infinite time interval

$$J = \int_0^\infty (x^T Q x + u^T R u) d\tau \tag{42}$$

System dynamics (LTI)

$$\dot{x} = Ax + Bu \tag{43}$$

with (A,B) stabilizable and $(A, Q^{1/2})$ detectable. It is possible to show that the corresponding Riccati equation

$$-\dot{P}(t) = P(t)A + A^T P(t) + Q - P(t)BR^{-1}B^T P(t)$$

boundary of $P(t)$ is limited

$$\lim_{T \to \infty} P(T) = 0 \tag{44}$$



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As $(t \to -\infty)$, this solution tends to a constant symmetric positive definite matrix, which can be found by solving the algebraic Riccati equation (ARE)

$$PA + A^T P - PBR^{-1}B^T P + Q = 0 \tag{45}$$

with the LQR-optimal control policy in a feedback form

$$u = -\underbrace{R^{-1}B^T P}_{K} x = -Kx \tag{46}$$

where $K \in \mathcal{R}^{m \times n}$ is a constant matrix of the LQR-optimal feedback gains.



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Steps for the Optimal Control in Infinite Horizon

- Dynamics

$$\dot{x} = Ax + Bu, x(0) = x_0 \tag{47}$$

- Performance index

$$J = \int_0^\infty (x^T Q x + u^T R u) d\tau \tag{48}$$

- Algebraic Riccati Equation

$$PA + A^T P - PBR^{-1}B^T P + Q = 0 \tag{49}$$

- Optimal Control

$$u = -R^{-1}B^T P x = -Kx \tag{50}$$

- Closed - Loop System

$$\dot{x} = (A - BK)x, x(0) = x_0 \tag{51}$$

Example

Consider the system defined by

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \tag{52}$$

The performance index J is given by

$$J = \int_0^\infty (x^T Q x + u^T R u) dt \tag{53}$$

where $Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $R = 1$ the state-feedback gain matrix K is offered for the controller

$$u = -Kx$$

Determine the optimal feedback gain matrix K

Example

With the procedures provided, check the controllability and detect-ability of the system matrix pairs (A,B) and (A, $Q^{1/2}$). And the Riccati equation , with a positive-definite matrix P, can be offered below

$$A^T P + PA - PBR^{-1}B^T P + Q = 0 \tag{54}$$

the result of P is

$$P = \begin{bmatrix} 0.4890 & 0.1481 \\ 0.1481 & 4.3014 \end{bmatrix}$$

substituting the solution of P into the following K, such that

$$K = R^{-1}B^T P = [1] \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} = [0.1481 \ 4.3014]$$



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Example

Consider system below

$$\dot{x} = Ax + Bu$$

where

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -35 & -27 & -9 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

and the performance index J is given by

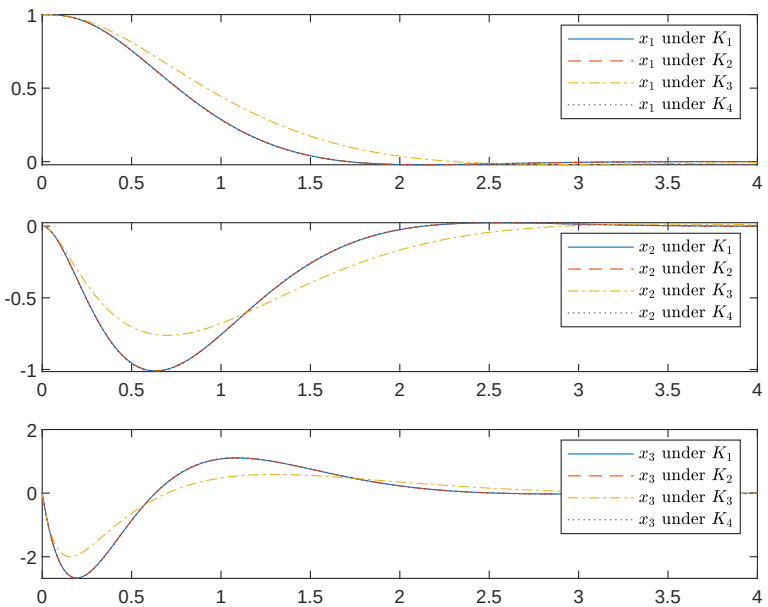
$$J = \int_0^\infty (x^T Q x + u^T R u) dt$$

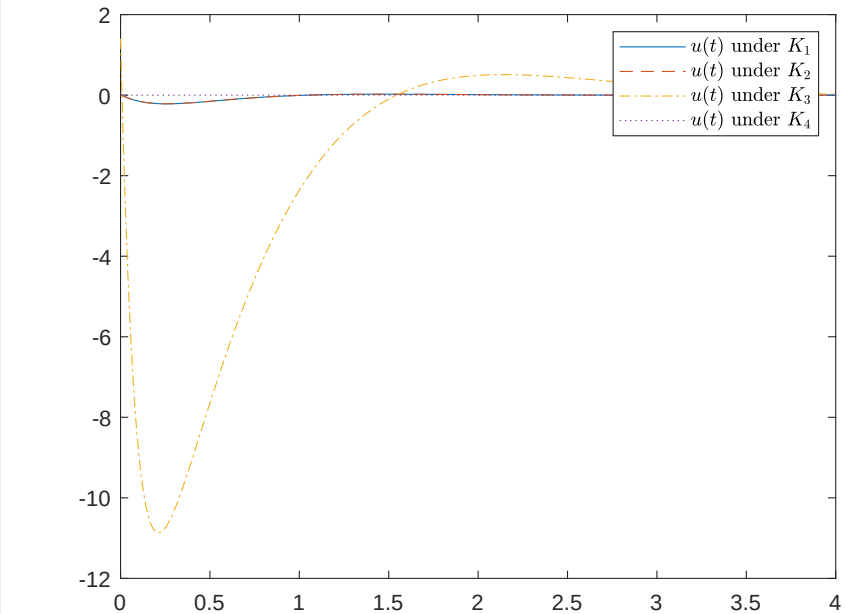
- Set $Q = I, R = 1$, calculate the optimal gain K_1;
- Set $Q = 100 * I, R = 100 * 1$, calculate the optimal gain K_2;
- Set $Q = 100 * I, R = 1$, calculate the optimal gain K_3;
- Set $Q = I, R = 100 * 1$, calculate the optimal gain K_4.

Example

Compare the resulting gains, the state responses and the control input. It can be obtained that

$$K_1 = [0.0143 \quad 0.1107 \quad 0.0676] \qquad K_2 = [0.0143 \quad 0.1107 \quad 0.0676]$$
$$K_3 = [1.4005 \quad 7.9353 \quad 5.0311] \qquad K_4 = [0.0001 \quad 0.0011 \quad 0.0007]$$





- With K_1 to K_2: Useless to amplify the parameters Q and R simultaneously. The ratio between Q and R is essential for the control gain;
- With K_1 to K_3: The larger the ratio Q to R, with means larger weight of system state performance, the gain is higher;
- With K_1 to K_4: The smaller the ratio Q to R, with means larger weight of system state performance, the gain is lower;
- With K_1 to K_2, K_1 to K_3: Higher ratio Q to R, will lead to higher control gain, and higher input consumption.

The balance and bargain between the parameters of the performance index should be adjusted according to the engineers' desire.

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4 Robust Control

- Model Uncertainties
- $H-\infty$ Control for Unknown Input

Consider a system with nonlinear uncertainties

$$\dot{x} = Ax + f(x) \tag{55}$$

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$$\dot{x} = Ax + f(x) \tag{55}$$

where $A \in \mathcal{R}^{n\times n}$ is the known nominal system parameter matrix. $f(x)$ is the perturbation satisfying

$$\frac{\|f(x)\|}{\|x\|} < \mu, \ \mu > 0 \tag{56}$$



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Theorem

The autonomous (zero input) system (55) satisfying the perturbation constraint (56) is stable, if there exists a positive matrix P with appropriate dimension that

$$PA + A^T P + 2\mu P < 0 \tag{57}$$

Proof.

Specifically, give a positive-definite Lyapunov function

$$V(x) = x^T P x \tag{58}$$

whose derivative

$$\begin{aligned} \dot{V}(x) &= 2x^T P \dot{x} \\ &= 2x^T P(Ax + f(x)) \\ &= 2x^T PAx + 2x^T Pf(x) \end{aligned} \tag{59}$$

with (56), one further obtains that

$$\begin{aligned} 2x^T Pf(x) &\leq 2\|x^T Pf(x)\| \\ &< 2\mu x^T Px \end{aligned}$$

such that,

$$\begin{aligned} \dot{V} &< 2x^T PAx + 2\mu x^T Px \\ &= 2x^T(PA + \mu P)x \end{aligned} \tag{60}$$

Therefore, if (57) holds, $\dot{V} < 0$ and the system (55) is stable. □

Example

Give the stability standard for the system

$$\dot{x} = Ax + Bu + f(x) \tag{61}$$

which satisfies the constraint (56), under the state-feedback controller

$$u = Kx \tag{62}$$

where gain matrix K is assumed to be known.
The system parameter A in (55) is to be replaced with

$$\bar{A} = A + BK$$



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4 Robust Control

- Model Uncertainties
- $H - \infty$ Control for Unknown Input

Consider a system in Fig.2
where z, y denote the system output and measured output signals. Variable z is the control output.

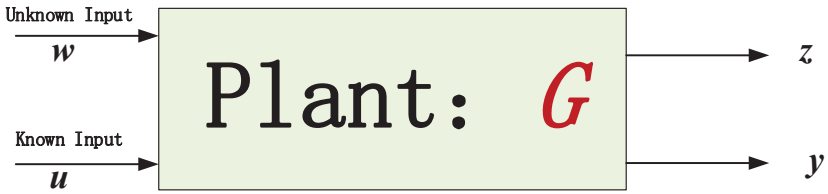


Figure 2: System with Known and Unknown Inputs

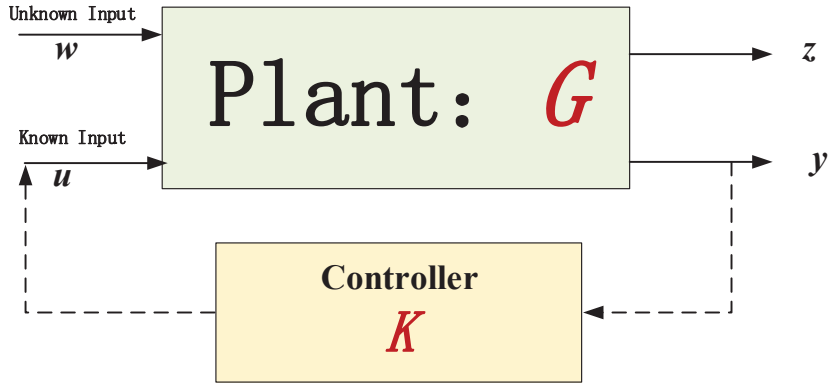


Figure 3: System under Robust Controller

The control target is to design controller as in Fig.3

$$u = Ky(t) \tag{63}$$

such that

$$\left\| \frac{z}{w} \right\|_{\infty} < \gamma, \quad \gamma > 0 \tag{64}$$



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the optimal aim is to minimize the effect of disturbance to the control output

$$\min \left(\left\| \frac{z}{w} \right\|_\infty \right) \tag{65}$$



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$$\|z\|^2 - \gamma^2 \|w\|^2 < 0 \tag{66}$$

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The measured output based closed-loop system can be depicted as

$$\begin{cases} \dot{x} &= (A + B_1 K C_2)x + (B_1 K D_2 + B_2)w \\ z &= (C_1 + D_{11} K C_2)x + (D_{11} K D_2 + D_{12})w \end{cases} \tag{67}$$



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The criterium on satisfying the requirement of (66) is that there exists a parameter $P>0$

$$\begin{bmatrix} M_1^T M_1 + N_1 + N_1^T & M_1^T M_2 + N_2 \\ * & M_2^T M_2 - \gamma^2 \end{bmatrix} < 0 \tag{68}$$

where $M_1 = C_1 + D_{11}KC_2$, $M_2 = D_{11}KD_2 + D_{12}$, $N_1 = P(A + B_1KC_2)$, $N_2 = P(B_1KD_2 + B_2)$

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Proof.

Define a Lyapunov function below

$$V(t) = x^T(t)Px(t) \tag{69}$$

whose derivative is

$$\begin{aligned} \dot{V}(t) &= 2x^T(t)P\left(\left(A + B_1KC_2\right)x(t) + \left(B_1KD_2 + B_2\right)w\right) \\ &= 2x^T(t)P\left(\left(A + B_1KC_2\right)x(t) + \left(B_1KD_2 + B_2\right)w\right) \\ &= \begin{bmatrix} x \\ w \end{bmatrix}^T \begin{bmatrix} P(A + B_1KC_2) + (A + B_1KC_2)^T P & P(B_1KD_2 + B_2) \\ * & 0 \end{bmatrix} \begin{bmatrix} x \\ w \end{bmatrix} \end{aligned} \tag{70}$$

□

Proof.

The optimal target (or cost function) can be verified as

$$J(u,\omega)=\int_{t_0}^{T}(z^Tz-\gamma^2\omega^T\omega)d\tau \tag{71}$$

where $\gamma\geq 0$ and with t_0, x_0 given and T, $x(T)$ free. The control goal is to find the optimal input u^* and maximizing disturbance such that

$$J(u^*,\omega)\geq J(u^*,\omega^*)\geq J(u,\omega) \tag{72}$$

one has that

$$\begin{aligned} J(u,x,\omega) &= \int_{t_0}^{T}\left(z^Tz-\gamma^2\omega^T\omega\right)d\tau \\ &= \int_{t_0}^{T}\left[\begin{array}{l}\left((C_1+D_{11}KC_2)x+(D_{11}KD_2+D_{12})w\right)^T\\ \times\left((C_1+D_{11}KC_2)x+(D_{11}KD_2+D_{12})w\right)-\gamma^2\omega^T\omega\end{array}\right]d\tau \end{aligned} \tag{73}$$

If one sets $M_1=C_1+D_{11}KC_2$, $M_2=D_{11}KD_2+D_{12}$, it has □

Proof.

$$J(x,\omega)=\int_{t_0}^{T}\begin{bmatrix} x \\ w \end{bmatrix}^T\begin{bmatrix} M_1^TM_1 & M_1^TM_2 \\ * & M_2^TM_2-\gamma^2 \end{bmatrix}\begin{bmatrix} x \\ w \end{bmatrix}d\tau \tag{74}$$

With (74) and (70), one has that if the below inequation holds, the system is stable and the optimal H_∞ condition can be satified。

$$\begin{bmatrix} M_1^TM_1+P\left(A+B_1KC_2\right)+\left(A+B_1KC_2\right)^TP & M_1^TM_2+P\left(B_1KD_2+B_2\right) \\ * & M_2^TM_2-\gamma^2 \end{bmatrix}<0 \tag{75}$$

□

1 Introduction

2 Optimal Control and Hamiltion - Jabobian - Bellman Equation

3 Linear Quadratic Regulator

4 Robust Control

5 Chapter Conclusion



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- Hamilton-Jacobian-Bellman
- LQR
- Robust Control: Model Uncertainties and Unknown Inputs



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